Library Management — Project 1

Lending Policies Specification

1 Scope

This document specifies the lending policies that must be supported by the Library Management system.

The system must support:

1. Standard Lending Policy
2. Adaptive Lending Policy
3. Lending Policy Selection

The quantitative parameters defined in this specification must be externally configurable at setup time, without requiring changes to the source code.

2 Standard Lending Policy

The Standard Lending Policy defines the following rules:

2.1. A reader may have at most *MaxActiveLendings* (> 0) active lendings.

2.2. The lending period is *LendingPeriod* (> 0) days.

2.3. A lending may be renewed at most *MaxRenewals* (>= 0) times, each time for *RenewalPeriod* (> 0) days.

2.4. A reader with any overdue lending cannot start a new lending.

2.5. An overdue lending cannot be renewed.

3 Adaptive Lending Policy

The Adaptive Lending Policy adapts the lending period according to the number of active lendings of the reader.

3.1. A reader may have at most *MaxActiveLendings* (> 0) active lendings.

3.2. If the reader has fewer than *FirstThreshold* active lendings, the lending period is *LongPeriod* (> 0) days.

3.3. If the reader has at least *FirstThreshold* but fewer than *SecondThreshold* active lendings, the lending period is *StandardPeriod* (> 0) days.

3.4. If the reader has at least *SecondThreshold* active lendings, the lending period is *ShortPeriod* (> 0) days.

3.5. The following constraints apply:

0 < FirstThreshold < SecondThreshold < MaxActiveLendings

LongPeriod > StandardPeriod > ShortPeriod > 0

3.6. A lending may be renewed only if the resulting lending continues to satisfy the applicable conditions of this policy.

3.7. A reader with any overdue lending cannot start a new lending.

4 Lending Policy Selection

The Lending Policy Selection determines which lending policy applies to a reader.

4.1. The Standard Lending Policy applies by default.

4.2. The Adaptive Lending Policy applies when, during the previous *HistoryPeriod* (> 0) months, the reader:

4.2.1. has completed at least *MinCompletedLendings* (> 0) lendings;

4.2.2. has returned at least *MinOnTimeRatio* (0–100%) of those lendings on time; and

4.2.3. has no currently overdue lending.

4.3. The applicable lending policy must be re-evaluated whenever a new lending or renewal is requested.

5 Parameter Configuration

5.1. All quantitative parameters identified in this specification must be externally configurable at setup time.

5.2. Changing the value of a parameter must not require changes to the source code.

5.3. Configured parameter values must satisfy the constraints defined by the corresponding lending policy.